

Advanced Coding Practice

HW Problem #2

CSE4152
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Largest N numbers

Let M be an $N \times N$ integer matrix, where each element in M is distinct, ensuring that no two elements are equal. Propose an algorithm that efficiently identifies the N largest elements among the elements of M . Furthermore, provide a proof of the correctness of this algorithm.

Example

Input

3

5 12 9

1 8 13

7 2 10

Answer:

13

12

10

Largest N numbers – Guiding questions

- How can we find the maximum number?
- How can we find the next maximum number?
- Divide an 1D array into two 1D equal-sized arrays.
 - Find the maximum in each array. How can we determine the overall maximum?
 - After removing the overall maximum, must we examine every element in both arrays again to find the next largest number?

Position of k

Consider an $N \times N$ matrix of integers with the following properties:

- Each row is sorted in non-**decreasing** order from left to right.
- Each column is sorted in non-**increasing** order from top to bottom.

Propose an algorithm that, given such a matrix and an integer k , efficiently determines the position of k within the matrix. If k occurs multiple times, identify any one occurrence.

Example

Input

8

10 11 20 23 50

4 8 15 21 30

3 7 14 18 28

2 6 11 16 27

1 5 9 13 25

Answer:

(2, 2)